

Mohammad Hashemi

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EDUCATION

- **Emory University** Atlanta, USA
Ph.D. – Computer Science; GPA: 3.97/4 Aug. 2023 - Present
- **Shahid Beheshti University** Tehran, Iran
B.Sc. – Computer Engineering; GPA: 3.85/4 Sep. 2018 - Feb. 2023

RESEARCH INTERESTS

Machine Learning, Spatial Computing, Graph Data Mining, Computer Vision

EXPERIENCE

- **Graduate Research Assistant** Jan. 2025 - Present
Spatial Computing Lab *Ph.D. Advisor: Prof. Andreas Züfle*
 - **Geospatial Agentic AI**: Agentic AI for geospatial data, Spatio-Temporal LLM agents
 - **Geospatial Representation Learning**: Understanding human mobility and POI data by graph representation learning
- **Graduate Research Assistant** Aug. 2023 - Dec. 2024
MELODY Lab *Supervisor: Dr. Wei Jin*
 - **Data-Centric AI**: Graph Data Reduction, Graph Condensation for GNNs
- **Graduate Teaching Assistant** Jan. 2024 - May. 2025
Computer Science Department at Emory University
 - **CS 470 - Data Mining**: Under the supervision of Dr. Wei Jin and Kai Shu
 - **CS 584 - Spatial Computing**: Under the supervision of Prof. Andreas Züfle
- **Undergraduate Research Assistant** Feb. 2021 - Feb. 2023
Image Processing & Distributed Systems Lab *B.Sc. Advisor: Prof. Mohsen Ebrahimi Moghaddam*
 - **Medical Image Computing**: Deformable 4D CT Image Registration, Medical Image Quality Assessment
- **Machine Learning Internship** May. 2021 - Aug. 2021
Rahnema College *Supervisor: Dr. Seyed Ali Osia*
 - **Web Crawler Detection**: Web crawler detection using unsupervised data mining
- **Undergraduate Research Assistant** Oct. 2020 - Jun. 2021
RoIAA Lab *Supervisor: Dr. Armin Salimi Badr*
 - **AI in Healthcare**: Time-series analysis for Parkinson disease prediction

PUBLICATIONS & MANUSCRIPTS (GOOGLE SCHOLAR LINK)

- **[TKDE] A Comprehensive Survey on Agentic AI for Geospatial Data:**
M. Hashemi, H. Amiri, A. Züfle, (Under Review)
- **[KDD 2026] Scalable Graph Condensation with Evolving Capabilities:**
M. Hashemi*, S.Gong*, J. Ni, C. Yang, W. Jin (* Equal Contribution)
- **[SIGSPATIAL 2026] PlaceRep: Geospatial Place Representation Learning from Large-Scale Point-of-Interest Graph Data:**
M. Hashemi, H. Amiri, A. Züfle
- **[SIGSPATIAL 2026] Staypoint Detection from Noisy Trajectory Data:**
L. Kennedy, H. Amiri, Y. Liu, R. Bao, H. Chen, M. Hashemi, R. Kong, X. Liu, J. Kim, S. Tang, L. Zhao, A. Züfle
- **[SIGSPATIAL 2026] Synthetic Wastewater Epidemiology Data Generation using Patterns-of-Life Simulation:**
H. Amiri, M. Hashemi, A. Deverakonda, J. Kim, Y. Wang, A. Züfle
- **[IEEE Data Descriptions] World-POI: Global Point-of-Interest Data Enriched from Foursquare and OpenStreetMap as Tabular and Graph Data:**
H. Amiri, M. Hashemi, A. Züfle
- **[IJCAI 2024] A Comprehensive Survey on Graph Reduction: Sparsification, Coarsening, and Condensation:**
M. Hashemi, S. Gong, J. Ni, W. Fan, B. Prakash, W. Jin
- **[Medical Physics] Automated Cardiac Coverage Assessment in Cardiovascular Magnetic Resonance Imaging Using an Explainable Recurrent 3D Dual-Domain Convolutional Network:**
S. Nabavi, M. Hashemi, M. Ebrahimi Moghaddam, A. Abin, A. Frangi
- **[Applied Computing & Geosciences] Machine learning Technique in the North Zagros Earthquake Prediction:**
S. Ommi, M. Hashemi

- **[JICSE 2023] Early Parkinson's Disease Diagnosis Based on Sequence Analysis:**
A. Salimi Badr, **M. Hashemi**
- **[Applied Intelligence] A Type-2 Neuro-Fuzzy System with a Novel Learning Method for Parkinson's Disease Diagnosis:**
A. Salimi Badr, **M. Hashemi**, H. Saffari
- **[IKT 2020] A Neural-Based Approach to Aid Early Parkinson's Disease Diagnosis:**
A. Salimi Badr, **M. Hashemi**

HONORS AND AWARDS

- Best B.Sc. thesis project in Computer Engineering - 2023
- 4th place, out of 131 undergraduate students studying Computer Engineering - 2020

SKILLS

- **Agentic AI & LLMs:** LLM Fine-tuning, Prompt Engineering, Tool Use, RAG, LLM-based Reasoning
- **Programming:** Python, C++, SQL, Git, Linux, GPU Computing, Distributed Training
- **ML Frameworks & Programming:** PyTorch, JAX, Scikit-learn, NumPy, Pandas, SciPy, OpenCV

PROFESSIONAL SERVICES

- **External Reviewer**
Conferences and Journals
 - **TKDE:** IEEE Transactions on Knowledge and Data Engineering - 2026
 - **SIGSPATIAL:** The Conference on Advances in Geographic Information Systems - 2025
 - **ICPR:** International Conference on Pattern Recognition - 2024
 - **AAAI:** AAAI Conference on Artificial Intelligence - 2024
 - **ICLR:** The International Conference on Learning Representations - 2024
- **Volunteering Experiences**
Conferences
 - **Web Chair:** WWW'24 Workshop in Data-centric Artificial Intelligence (Website) - 2024